CSI 2132 – Databases I

- Course Overview

January 2026, Verena Kantere

Taught by

Professor: Verena Kantere

Office: SITE 5 005

Email: vkantere@uottawa.ca

Research Specialization: Data Management, Databases, Big Data, Cloud computing, IoT systems, Data Analytics, Information Systems

I have been teaching courses on Databases I, II and Advanced Databases for over 16 years.

Teaching Assistants and correctors

There are 8 TAs for the course: 4 for Section A and 4 for Section B.

There are 2 correctors for the course: 1 for each section.

All TAs will be involved in delivering lab and tutorials. They will help students with questions on the material of the course. Each one of them will be mostly responsible for questions on specific parts of the examination and respective course material:

Section A:

- (TA) Harsh Jha, hjha091@uottawa.ca : responsible for the midterm & project deliverables 1
- (TA) Parna Asadi pasad086@uottawa.ca: responsible for the course project deliverable 2
- (TA) Minoo Soltanshahi msolt085@uottawa.ca : responsible for assignment 1 & assignment 2
- (TA) Yi Chen ychen808@uottawa.ca : responsible for assignment 3
- (Corrector) Patrick Meyer pmeye103@uottawa.ca : responsible for the correction of the midterm and final exam

If you need help with a specific part of the material or assignment you can contact the responsible TA.

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Section B:

- (TA) Arham Farooq Mohammed amoha058@uottawa.ca : responsible for project deliverable 1 & assignment 3
- (TA) Angelo Ayoub aayou086@uottawa.ca : responsible for assignment 1
- (TA) Farnaz Jazaeri fjaza018@uottawa.ca : responsible for assignment 2
- (TA) Sina Negarandeh, snega084@uottawa.ca : responsible for project deliverable 2
- (Corrector) Shruthi Perumalsamy Gurunathan speru037@uottawa : responsible for the correction of the midterm and final exam

If you need help with a specific part of the material or assignment you can contact the responsible TA.

Taught at

The course is delivered in-person. The lectures are:

Section A:

- Monday 19:00-21:50: from 12 January to 06 April ? 2026
- Classroom: 161 Luis Pasteur (CBY) C03

Section B:

- Tuesday 19:00-21:50: from 13 January to 07 April 2026
- Classroom: 120 University (FSS) 2005

Teaching mode

The course is not bimodal:

- In-person lectures/labs/tutorials will NOT be transmitted online.
- Online delivery of the lecture might be chosen in case of instructor's illness or a snowstorm.
- In case of online delivery of the lecture:
  - The lecture will be delivered online for all students and not both in-person and online.
  - Such a modification of lecture delivery will be announced the same day of the lecture by 15:00.
  - A lecture that is delivered online will be recorded and the recording will be available in Brightspace for around 4 weeks.

Taught at

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Class	Class Title	Enrolled	Days & Times	Room	Class Dates
CSI 2132-A00 (1827)	DATABASES I (Lecture)	135	Mo 7:00PM - 9:50PM	161 Louis Pasteur (CBY) C03	Jan 12, 2026-Apr 15, 2026
CSI 2132-A01 (1828)	DATABASES I (Laboratory)	33	Mo 4:00PM - 5:20PM	800 King Edward (STE) 2060	Jan 12, 2026-Apr 15, 2026
CSI 2132-A02 (1829)	DATABASES I (Laboratory)	15	Mo 4:00PM - 5:20PM	800 King Edward (STE) 2060	Jan 12, 2026-Apr 15, 2026
CSI 2132-A03 (1830)	DATABASES I (Laboratory)	38	Mo 5:30PM - 6:50PM	800 King Edward (STE) 0130	Jan 12, 2026-Apr 15, 2026
CSI 2132-A04 (2179)	DATABASES I (Laboratory)	34	We 5:30PM - 6:50PM	800 King Edward (STE) 0131	Jan 12, 2026-Apr 15, 2026
CSI 2132-A05 (1831)	DATABASES I (Laboratory)	15	Fr 5:30PM - 6:50PM	800 King Edward	Jan 12, 2026-Apr 15, 2026
CSI 2132-A06 (2178)	DATABASES I (Tutorial)	110	Fr 4:00PM - 5:20PM	65 University (MRT) 212	Jan 12, 2026-Apr 15, 2026
CSI 2132-A07 (2292)	DATABASES I (Tutorial)	25	Fr 5:30PM - 6:50PM	65 University (MRT) 212	Jan 12, 2026-Apr 15, 2026
CSI 2132-B00 (1832)	DATABASES I (Lecture)	123	Tu 7:00PM - 9:50PM	120 University (FSS) 2005	Jan 12, 2026-Apr 15, 2026
CSI 2132-B01 (1833)	DATABASES I (Laboratory)	61	Mo 2:30PM - 3:50PM	161 Louis Pasteur (CBY) B02	Jan 12, 2026-Apr 15, 2026
CSI 2132-B02 (1834)	DATABASES I (Laboratory)	22	Mo 2:30PM - 3:50PM	161 Louis Pasteur (CBY) B02	Jan 12, 2026-Apr 15, 2026
CSI 2132-B03 (1835)	DATABASES I (Laboratory)	40	Mo 5:30PM - 6:50PM	800 King Edward (STE) 2052	Jan 12, 2026-Apr 15, 2026
CSI 2132-B06 (2181)	DATABASES I (Tutorial)	76	Mo 4:00PM - 5:20PM	125 University (MNT) 207	Jan 12, 2026-Apr 15, 2026
CSI 2132-B07 (2293)	DATABASES I (Tutorial)	47	Mo 4:00PM - 5:20PM	801 King Edward (MNO) E218	Jan 12, 2026-Apr 15, 2026

Main Content

Office Hours

- The professor's office hours are ONLINE 11:00-13:00 every Wednesday ONLY BY APPOINTMENT ONLINE:
  - Please contact the professor by email at the latest by 19:00 the Tuesday before the Wednesday that you would like to meet with her online. The online meetings will take place via MS Teams with 1-1 ad hoc meetings.
- The TAs office hours are 10:00-12:00 on some Mondays or Thursdays at SITE 5058 during the semester.
  - The office hours of TAs will be announced around a week earlier than they will occur.
- If you need to see the professor in person, this can be arranged by email, usually on a Thursday.

Course Page and MS Team

Brightspace link:

https://uottawa.brightspace.com/d2l/home/560807

MS Teams link:

https://teams.microsoft.com/l/team/19%3AmyRpgnd6q-28ClCWicE7XKXJvrfSEqWVRt06eAA5GlU1%40thread.tacv2/conversations?groupId=3c5167af-041a-420a-9b65-13317998c227&tenantId=d41fdab1-7e15-4cfd-b5fa-7200e54deb6b

Yuja (from within Brightspace) will be used in case that the course needs to be delivered online.

Recordings of most lectures from last year are available in BrightSpace.

Suggested Bibliography

Main book:

- **Database System Concepts,** Silberschatz, Korth & Sudarshan
  https://www.db-book.com/
  https://codex.cs.yale.edu/avi/db-book/

Other well-known traditional books:

- **Fundamentals of Database Systems,** Elmasri & Navathe
  https://www.researchgate.net/publication/336472480_Fundamentals_of_Database_System
- **Database Systems: The Complete Book,** Hector Garcia-Molina, Jeff Ullman, and Jennifer Widom
  http://infolab.stanford.edu/~ullman/dscb.html
- **Database Management Systems,** Raghu Ramakrishnan and Johannes Gehrke (easily found on the web)
- **Database System Implementation,** Hector Garcia-Molina, Jeff Ullman, and Jennifer Widom (easily found on the web)
- **A First Course in Database Systems**, 3nd Edition, Jeff Ullman, and Jennifer Widom, (easily found on the web)

Course Goal and Curriculum

The goal of this course is to introduce the fundamentals of database management.

The course curriculum is:

1. Introduction to data management and databases*
2. Entity Relationship model
3. Relational model
4. Relational algebra and relational calculus
5. The query language SQL
6. Database normalization (normal forms and functional dependencies)
7. Physical organization and storage
8. Hashing and indexing

*The introduction to databases will not be examined. The purpose of it is to give to the students a good idea of what is traditional and modern data management and databases

Expected Outcomes of the Course

- A solid knowledge of the basic data management theoretical notions, issues and solutions:
  - Logical design of a relational database including conceptual design and correctness or appropriateness of the design
  - Basic query languages and how they compare
  - Physical storage techniques of relational data
  - Associative data structures: views and basic indexing schemes
- Hands-on experience with classical relational DBMSs:
  - Installing and operating a DBMS
  - Designing, implementing, and querying a database
- Creating a basic web application on top of a database

Evaluation

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Grading

- Midterm exam 10%
- Assignments 15%
- Course project 25%
- Final exam 50%

Requirements for success in the course:

- 50% or more of the grade in the final exam
- 50% or more of the grade in the course project

Course Assignments

There will be 3 short assignments:

1. (5%) Assignment on Relational Algebra and Relational Calculus
2. (5%) Assignment on Normalization and Functional Dependencies
3. (5%) Assignment on Indexing and Hashing

- The assignments should be submitted by each student individually
- **The assignment solutions will NOT be announced**.
- However, the students will receive individual feedback on their solutions and will be able to meet in-person or online with the TAs to discuss their solutions
- We will consider presenting selectively assignment solutions during some tutorials (this will be announced). In such a case you will be able to make notes, but no relevant material will be handed out.

Course Project

In teams (i.e. 1-3 students together) you will have to design and implement a database and develop a web application on top of it, for which we will give you the requirements.

You will have to submit:

1st part:

1. The ER diagram
2. The relational database schema (with declaration of all constraints)

2nd part:

1. The implemented database
2. The web application
3. Queries/triggers/views etc on the database needed for the functionality of the database

Course Project Deliverables

You will have to submit:

1st part:

1. The ER diagram
2. The relational database schema (with declaration of all constraints)

- This will be a deliverable in pdf format, one deliverable per team.

2nd part:

1. The implemented database
2. The web application
3. Queries/triggers/views etc on the database needed for the functionality of the database

-This will be a deliverable (one deliverable per team) that includes:

a. A report in pdf format

b. A link to a repository with your code

c. A short video of around 15 minutes in which you will explain your answer to all the requirements (instructions about the video will be given)

Course project - practicalities

- A student project can be submitted by teams of 1-3 students (the team members may come from both course sections).
- A deadline will be announced for you to declare your team members.
- The team composition cannot change after its declaration.
- You should try to find teammates and not do the project alone:
  - The project is substantial work for one student. Except if you already have knowledge of web app development and databases, you are advised to find other students to divide the work.
  - One major goal of the project is that students practice in collaborating with others. This simulates how IT project are developed in the industry.
  - Collaboration has to do with estimating the amount of work, the type of work, appropriately assigning parts of the work to the person that is most qualified and most motivated to do it, create a timeline for internal milestones and deliverables, devising appropriate 'interfaces' in order to allow people to work in parallel

Midterm Exam

- The midterm will take place in class instead of a lecture:
  - **Monday/Tuesday, 23-24 February, 2026**

- The duration of the midterm is 1 hour.
- The midterm exam is a written exam.
- The midterm exam is conducted with closed books and notes.
- <u>The midterm will examine knowledge on the ER model and the Relational Model.</u>

The midterm is not mandatory. If a student does not take the midterm, the 10% of this exam will be added to the final exam. However, the midterm is easy! So, don't avoid it!

Final Exam

- The goal of the final exam is to evaluate if the student has understood the theoretical notions and query languages and techniques discussed in the course.
- The final exam may include:
  - questions of critical thinking and comprehension of the material.
  - exercises of gradual difficulty that can be solved based on the taught material.
- The final exam is a written exam.
- The final exam is conducted with closed books and notes.

Tentative Timeline of the Course in 2026

12-13 Jan	First lecture
26 Jan	Project announcement
03 Feb	Assignment 1 announcement
23-24 Feb	Midterm exam (1.5 hours – the rest is lecture)
25 Feb	Submission of project deliverable 1 (part 1)
02 Mar	Assignment 1 submission
10 Mar	Assignment 2 announcement
30 Mar	Assignment 2 submission
31 Mar	Assignment 3 announcement
07 Apr	Submission of project deliverable 2 (part 2)
12 Apr	Assignment 3 submission

Following the Course

- Please come to all lectures, tutorials and labs
  - Make questions in class, communicate with me and the TAs
- The TAs are here to assist you constantly!
  - Ask the TAs first and not the professor: there are more than 260 students in both course sections and it is impossible for the professor to answer emails or messages from too many students.
- Do not leave work on assignments and especially the project for the last minute!
  - Do not send submissions via email – there will be not reply to such emails and such submissions will not be evaluated.
  - Submission sites will be left open for 4 days after the deadline in order to accommodate late submissions
  - If you submit late, please accompany your submission with a comment justifying the delayed submission – do not send an email.
  - If you are unable to submit even after 4 days, you will need to provide adequate official documentation that justifies the delay (e.g. medical certificate).

Course Evaluation

- Please evaluate the course!
- Please evaluate the course responsibly and not based on whether the course is easy or difficult!
  - This course teaches a fundamental part of computer science. It is a challenging course because:
    - It contains substantial theoretical material (the course is NOT only about learning SQL!).
    - It contains a hands-on extended project.
    - The project requires that you also learn the basics of web app development (which is not part of the official course curriculum). We make our best to ensure that you get all necessary knowledge in order to succeed with the web app development.

I would be happy to discuss ANY issue related to the course, but also to your studies, or beyond.

I wish you a happy semester!